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Corte, Jr. et al.

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(54) **DOUBLE PIECE RETAINER FOR COMPACT VALVE**

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This patent is subject to a terminal disclaimer.

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(22) Filed: **May 31, 2024**

Related U.S. Application Data

(63) Continuation of application No. 17/857,231, filed on Jul. 5, 2022, now Pat. No. 12,000,506, which is a continuation of application No. 16/931,957, filed on Jul. 17, 2020, now Pat. No. 11,378,190, which is a continuation of application No. 16/113,050, filed on Aug. 27, 2018, now Pat. No. 10,718,437, which is a continuation of application No. 14/631,970, filed on Feb. 26, 2015, now Pat. No. 10,060,538.

(60) Provisional application No. 61/946,252, filed on Feb. 28, 2014.

(51) **Int. Cl.**
F16K 5/06 (2006.01)

(52) **U.S. Cl.**
CPC **F16K 5/0642** (2013.01); **F16K 5/0626** (2013.01); **F16K 5/0689** (2013.01); **F16K 5/06** (2013.01); **Y10T 137/0508** (2015.04); **Y10T 137/6035** (2015.04); **Y10T 137/6041** (2015.04)

(58) **Field of Classification Search**

CPC F16K 5/0642; F16K 5/0626; F16K 5/0689;
F16K 5/06; F16K 27/067; Y10T
137/0508; Y10T 137/6035; Y10T
137/6041

See application file for complete search history.

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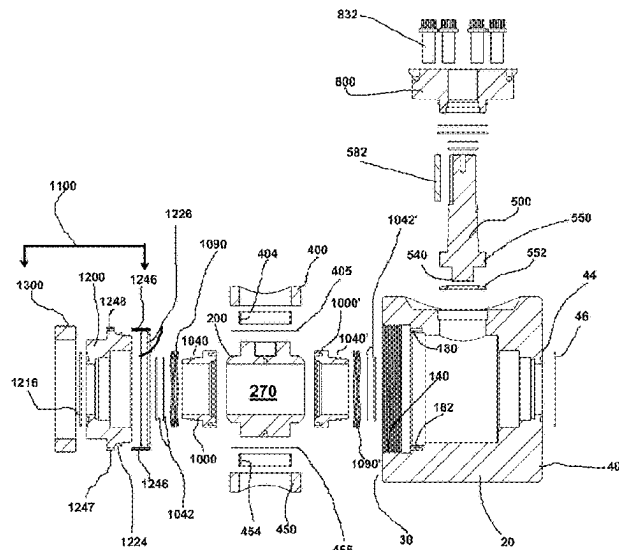
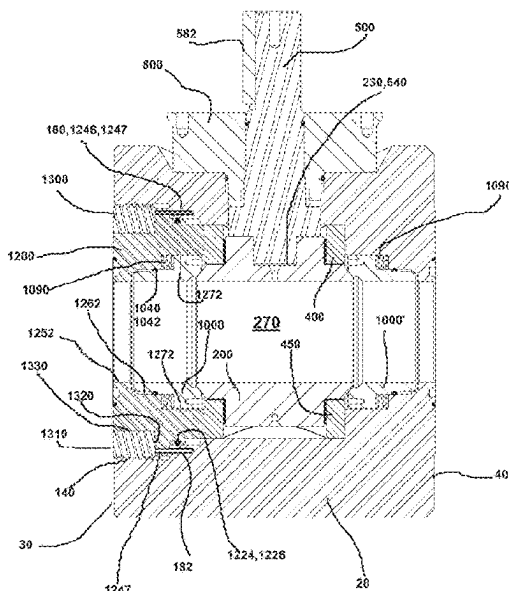
Primary Examiner — Hailey K. Do

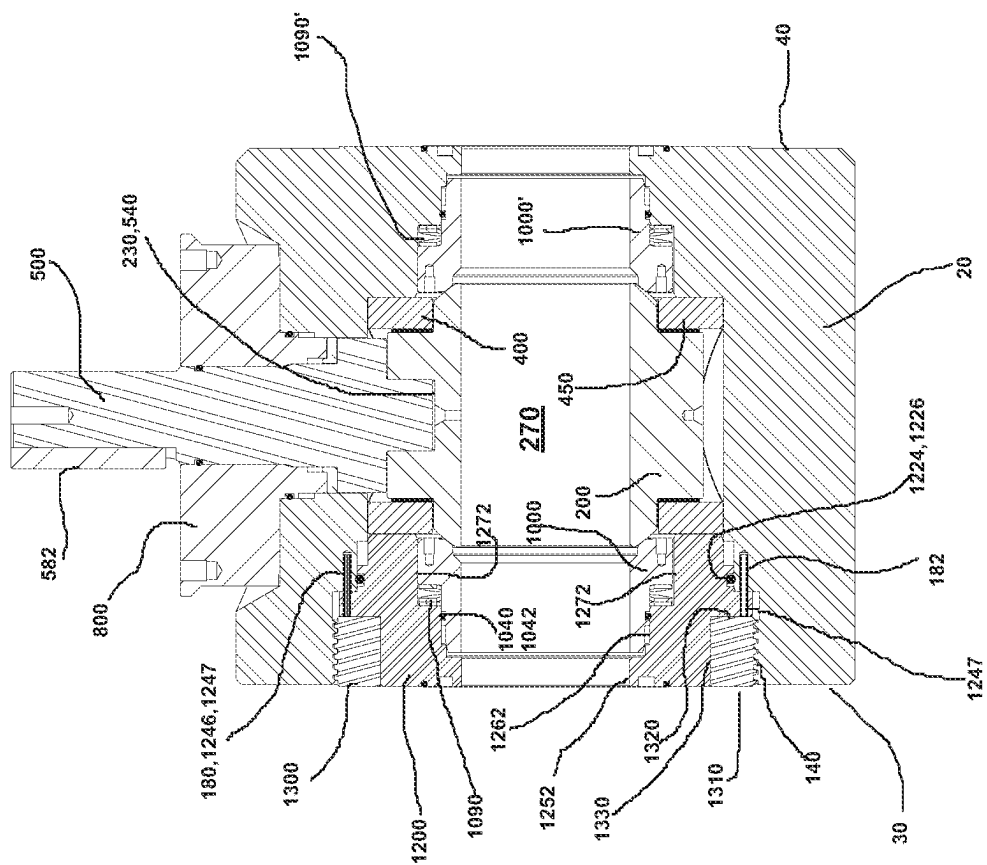
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& North, APLC; Brett A. North

(57) **ABSTRACT**

A compact manifold ball valve having a valve body, valve ball, valve stem and two piece retainer system; the valve body including a valve chamber having first and second ends, and a first flow passage intersecting the valve chamber and valve ball. The valve ball can be trunnion supported and include pair of movable seat assemblies that can be used to seal the valve ball to the valve body. A two piece retainer can be used to hold in place the valve components, the two piece retainer including a first section which moves only linearly along with a second retainer piece that is a threaded ring that rotationally locks in place the first section of the two piece retainer. The two pieces of the retainer can be symmetrically located about the centerline of the flow passage of the valve body.

20 Claims, 16 Drawing Sheets





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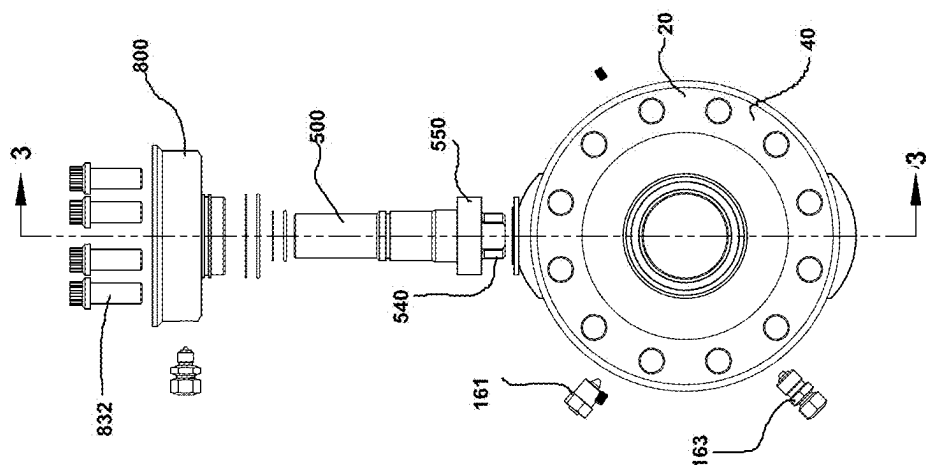


FIG. 2

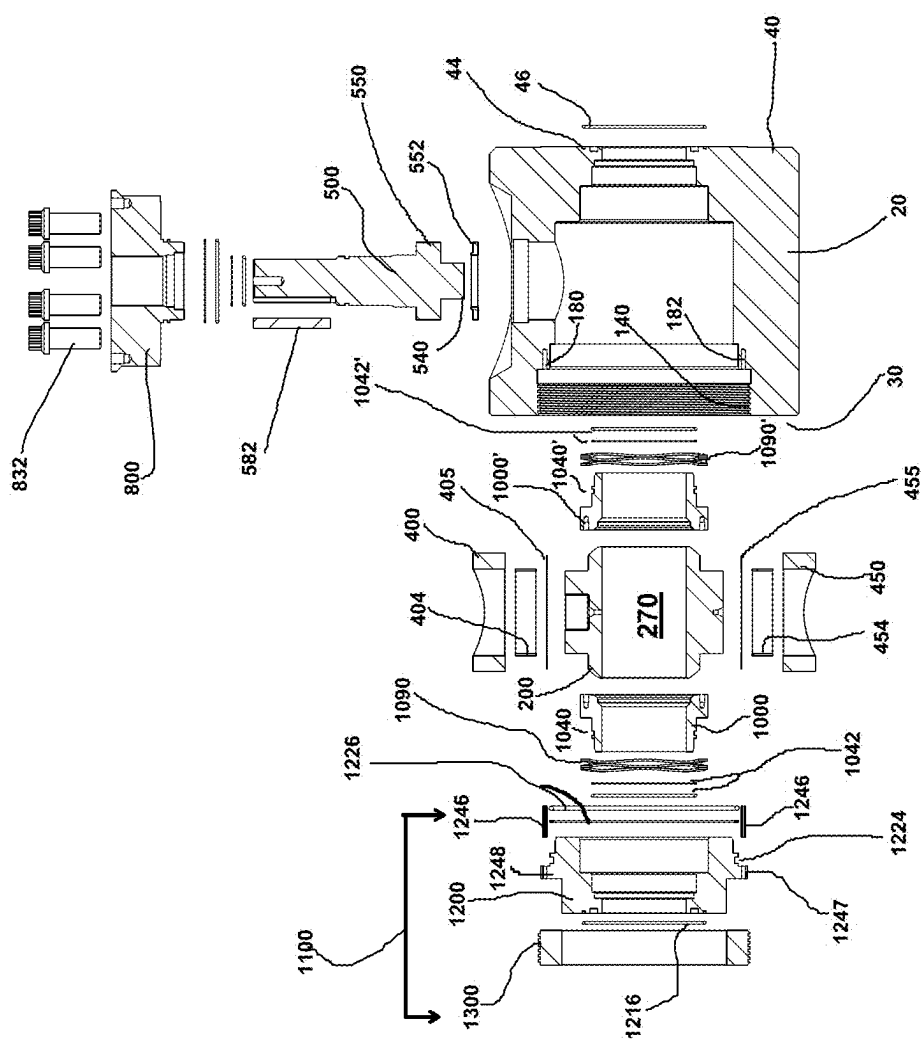


FIG. 3

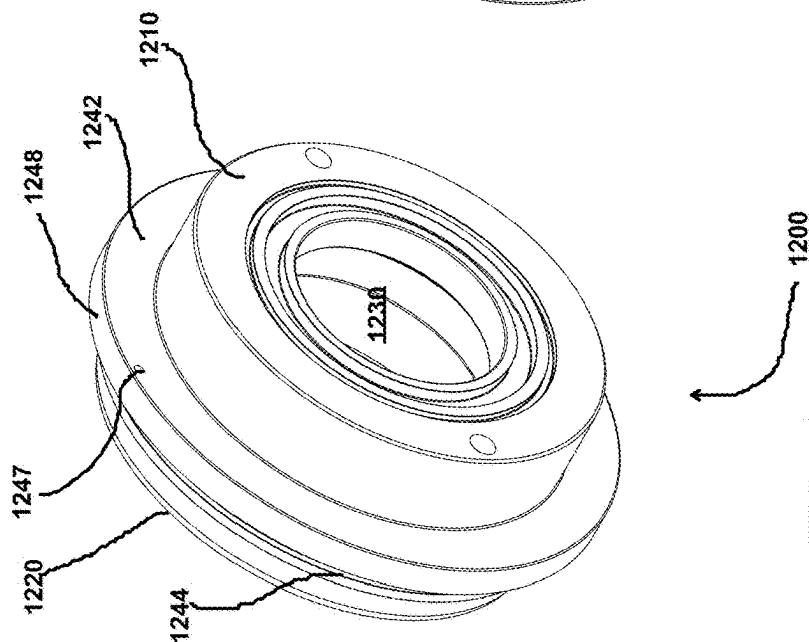


FIG. 4

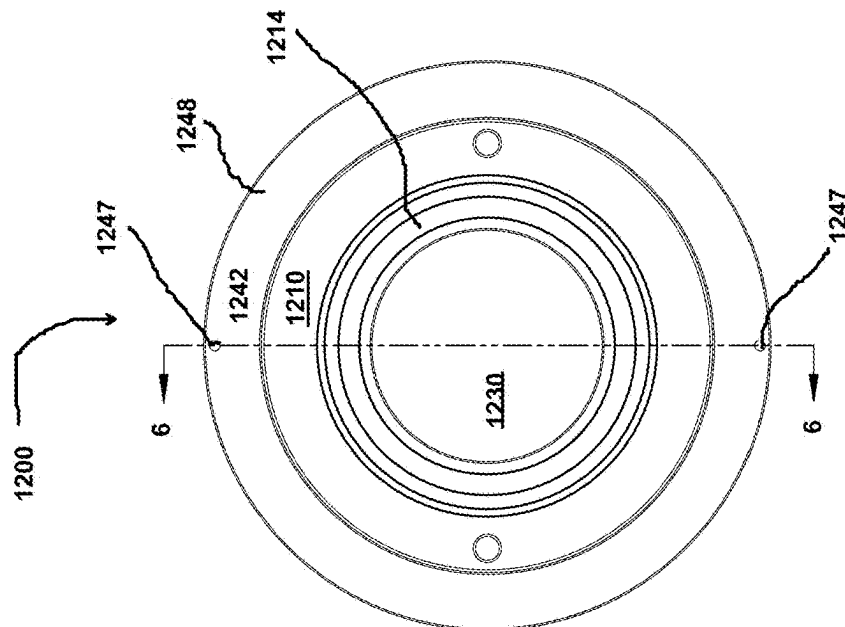


FIG. 5

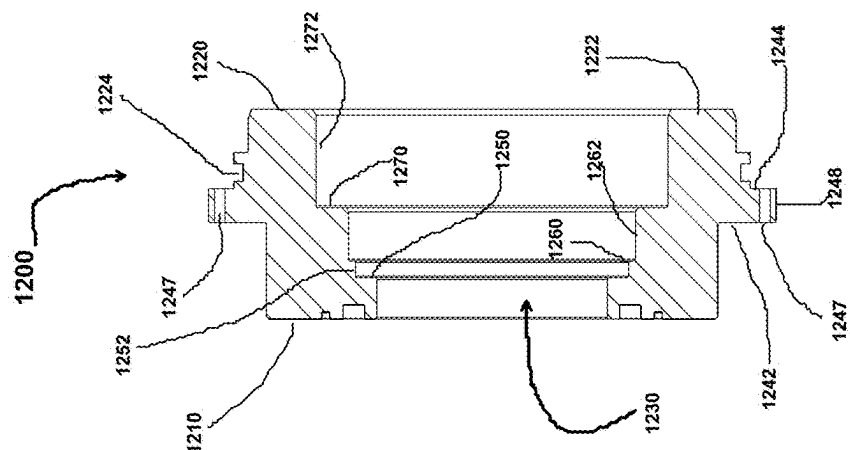


FIG. 6

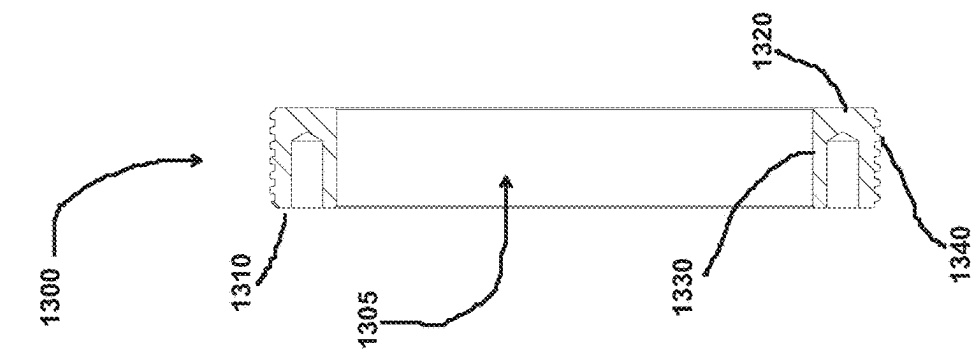


FIG. 7

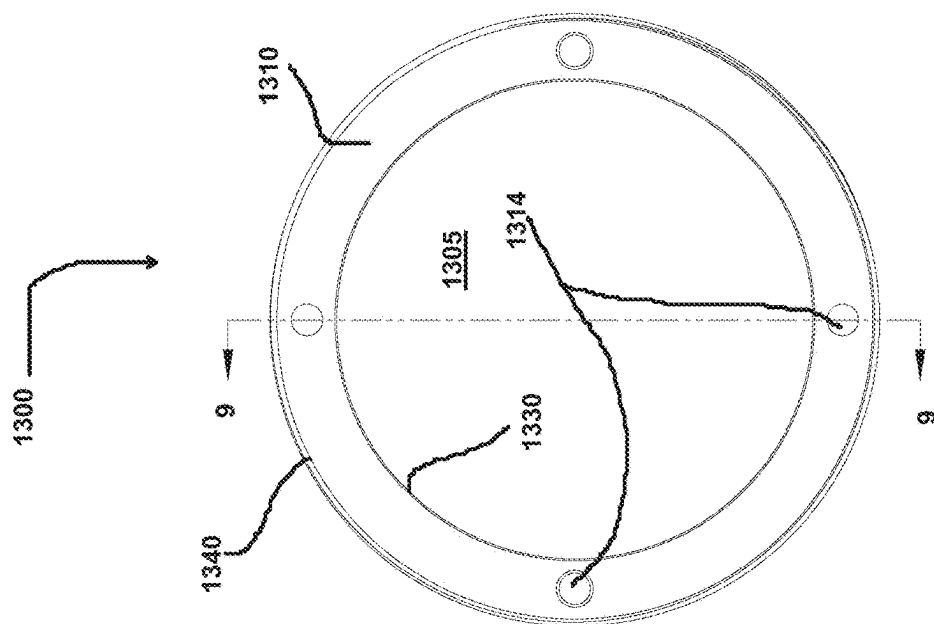


FIG. 8

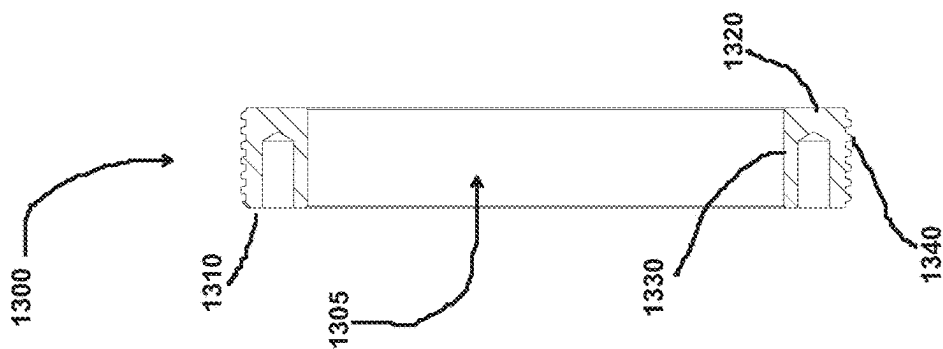


FIG. 9

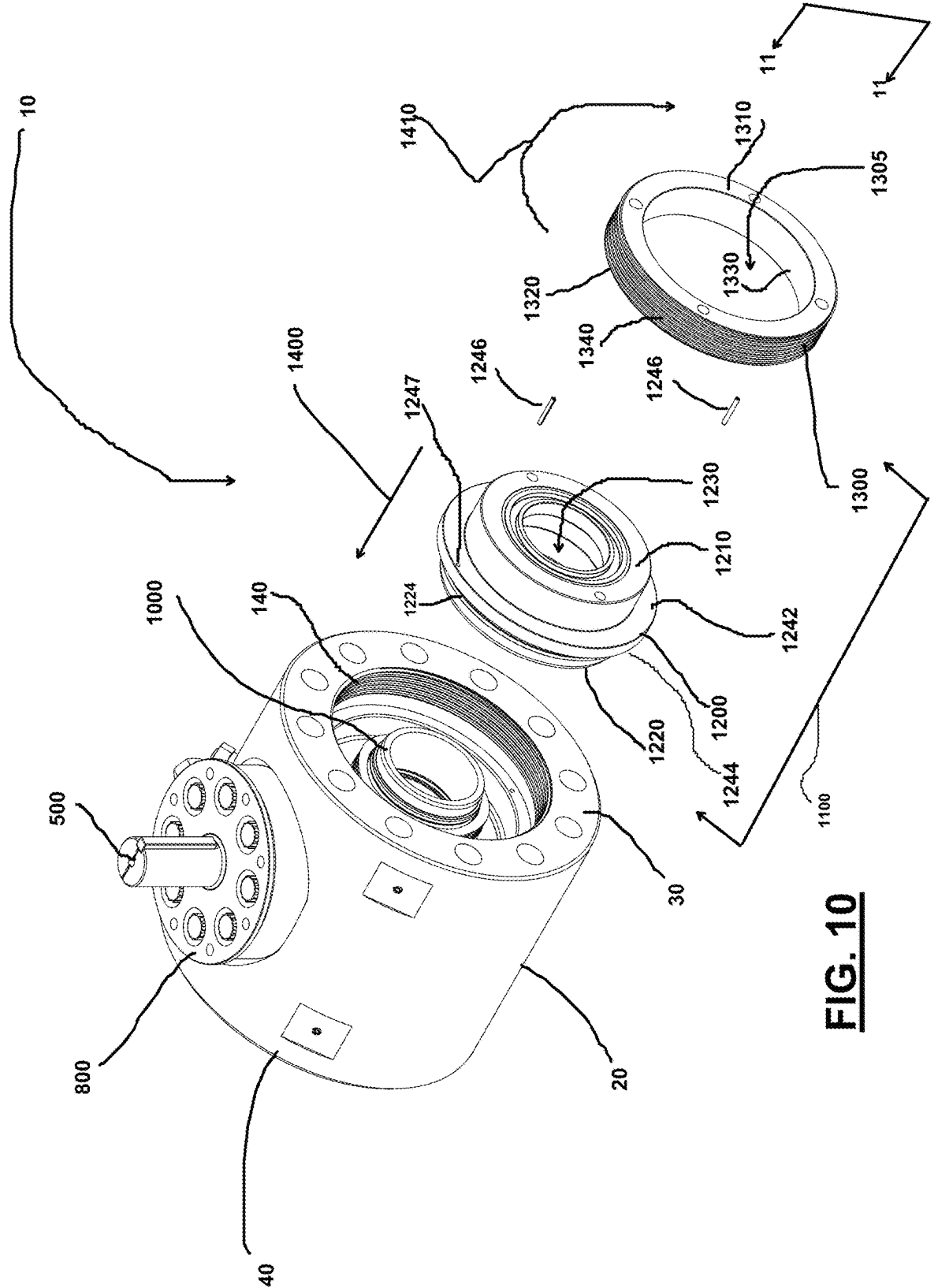
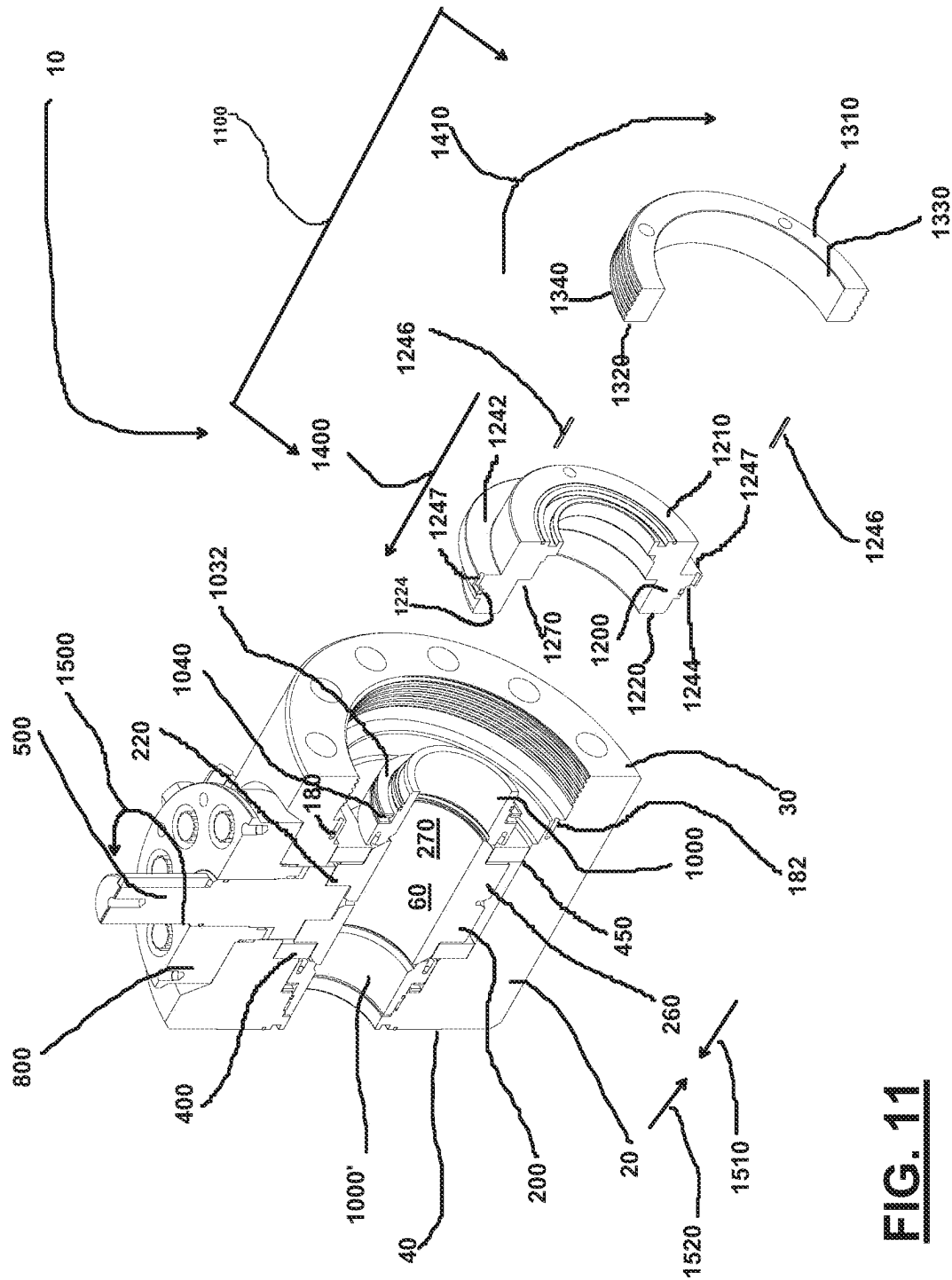


FIG. 10



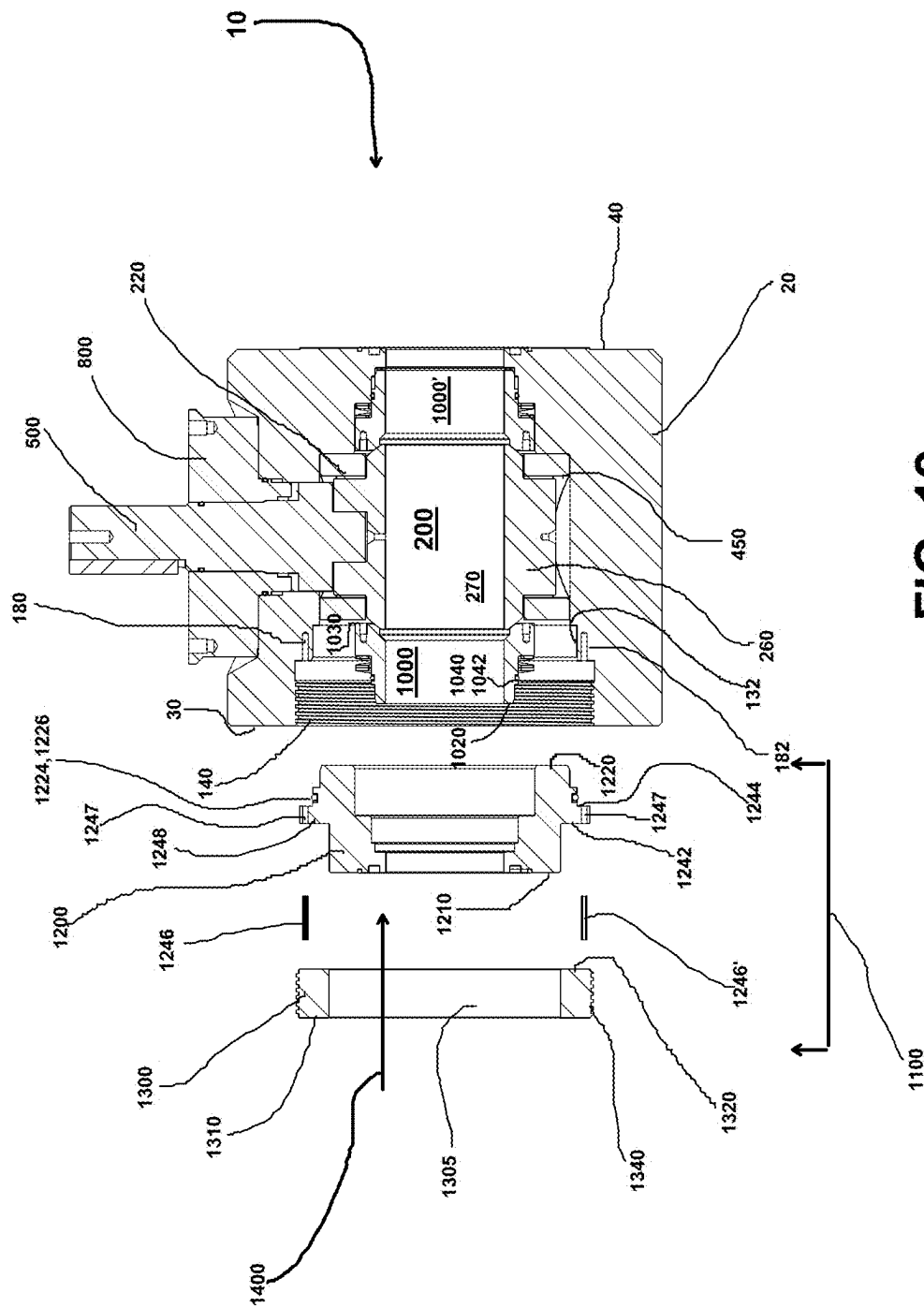


FIG. 12

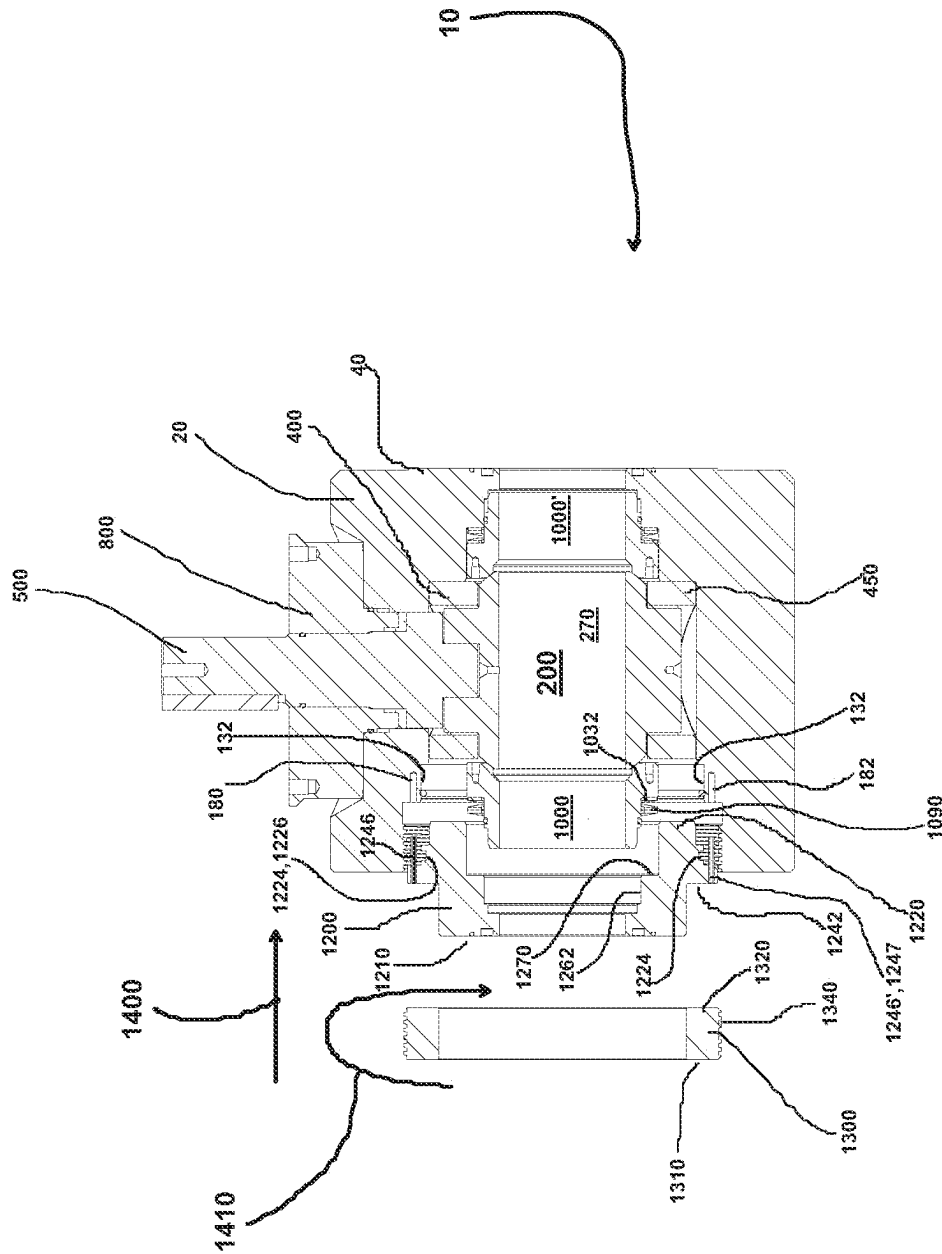


FIG. 13

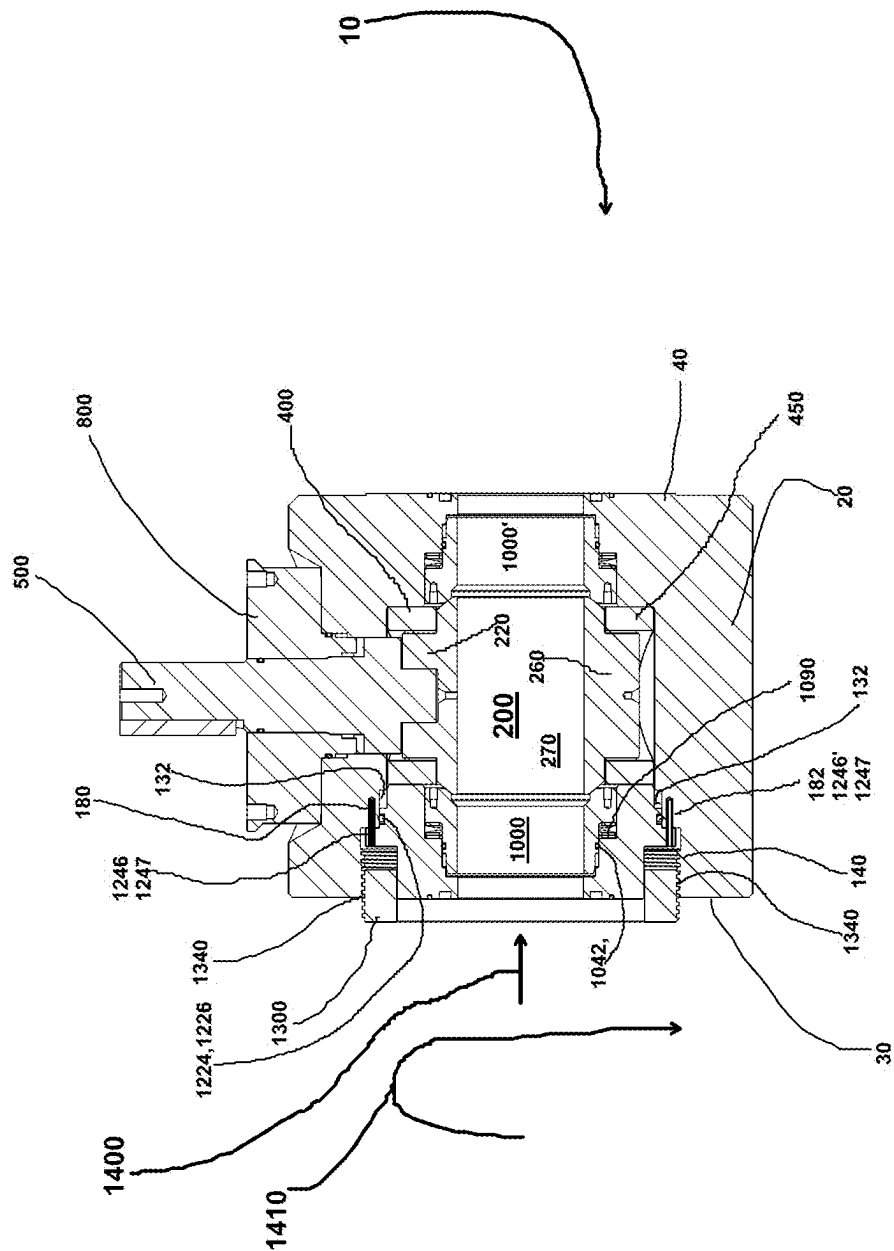


FIG. 14

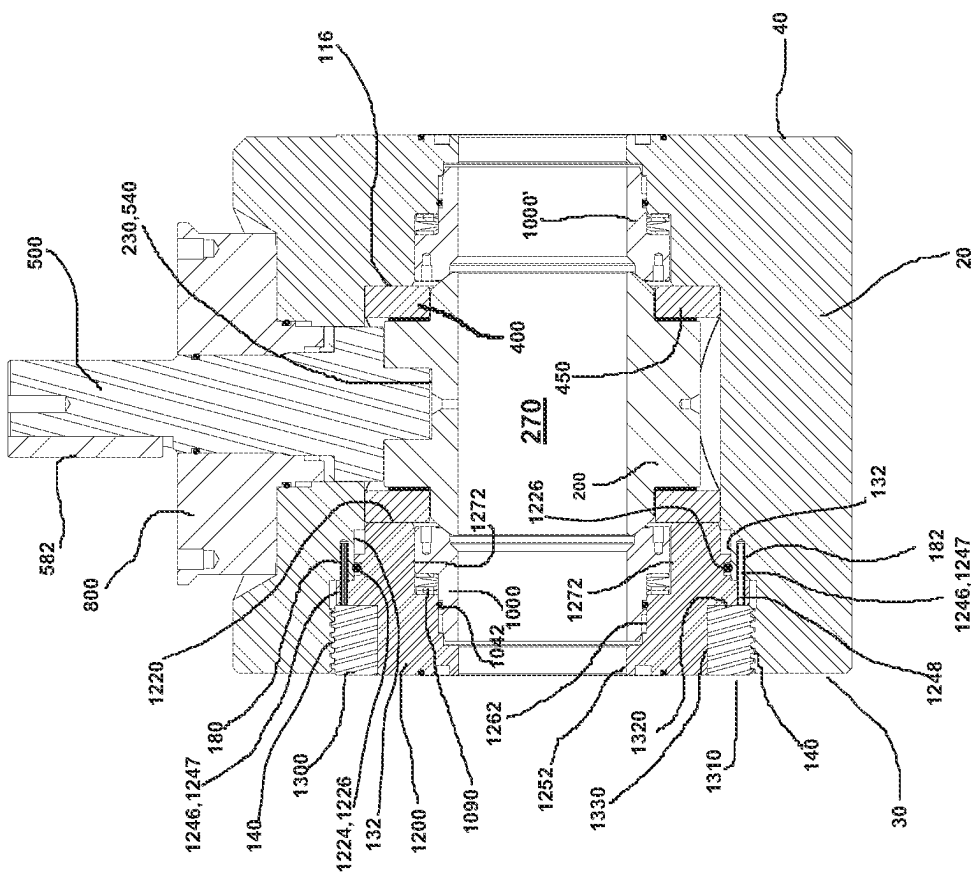
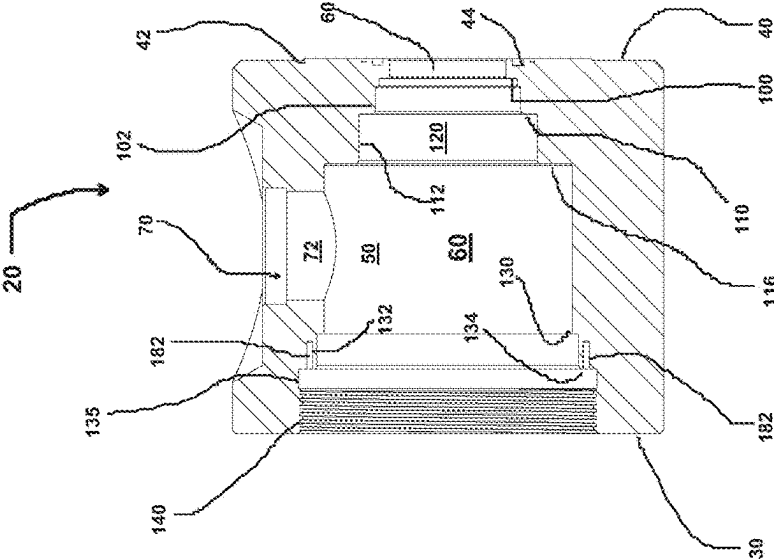
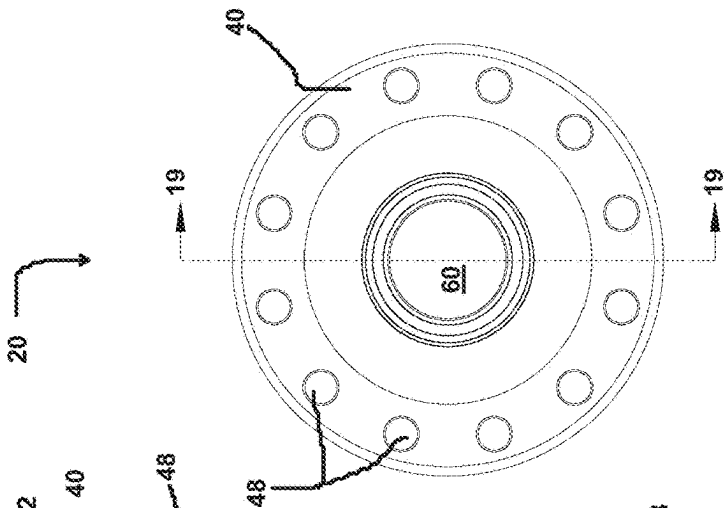
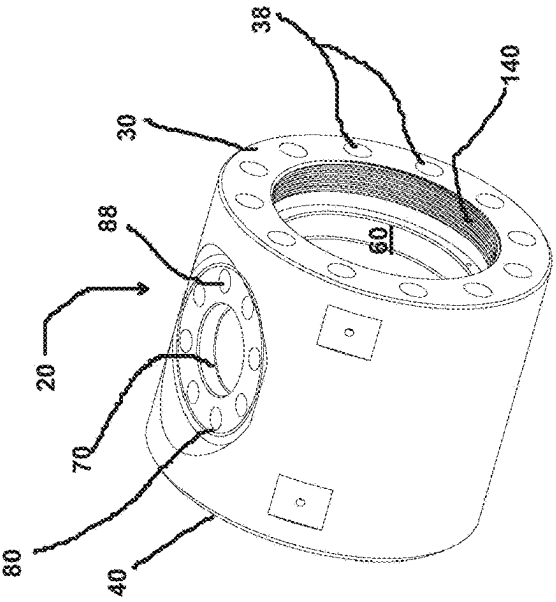
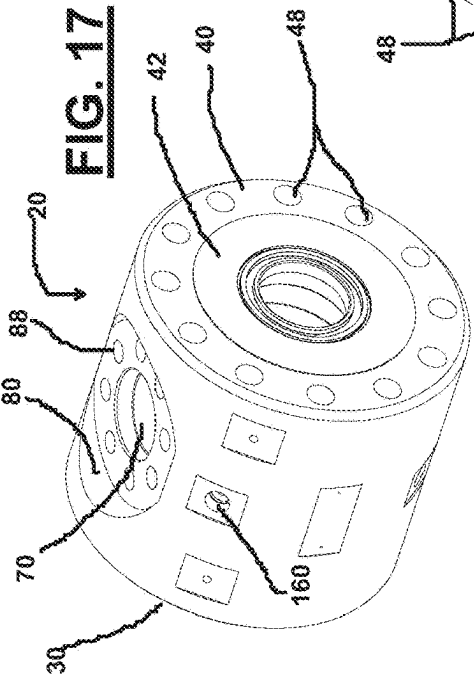
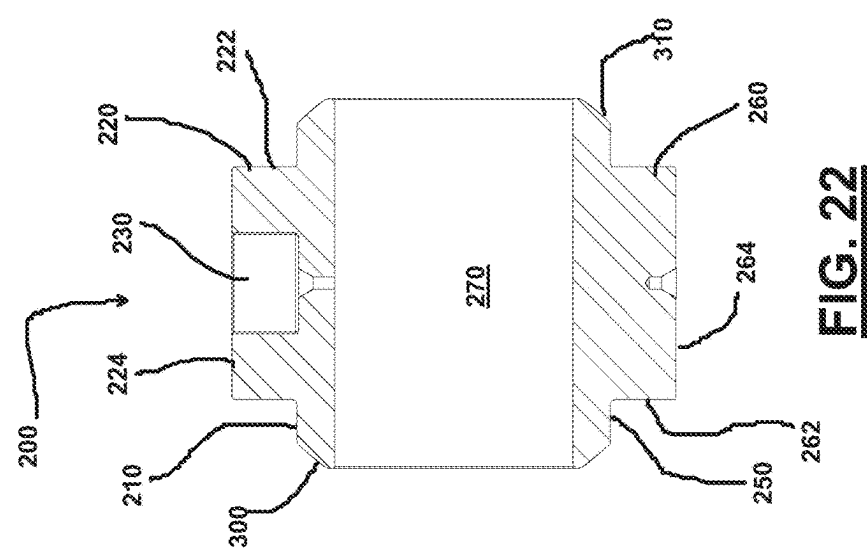
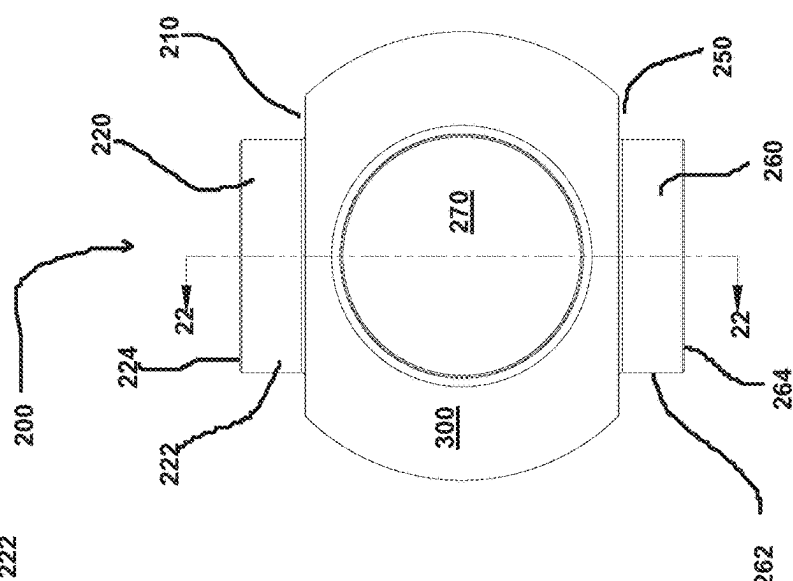
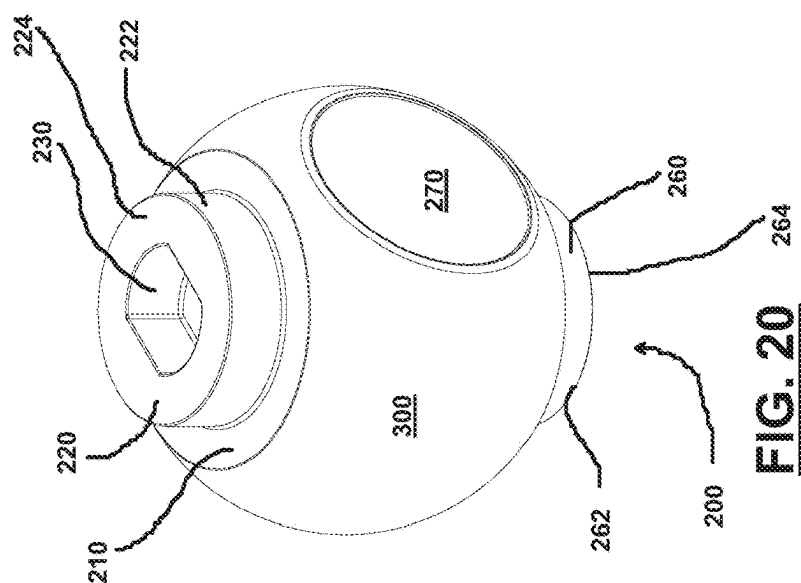


FIG. 15





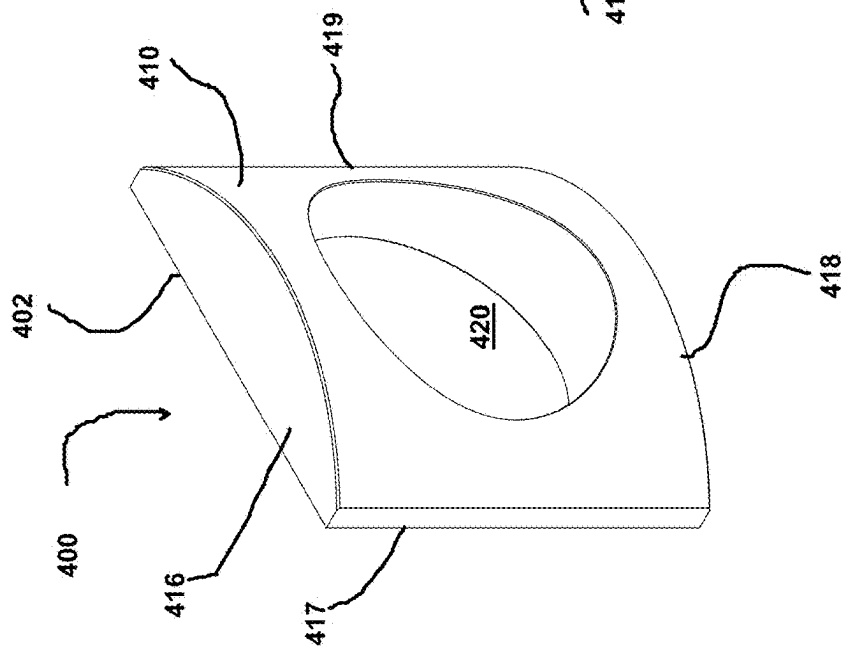


FIG. 23

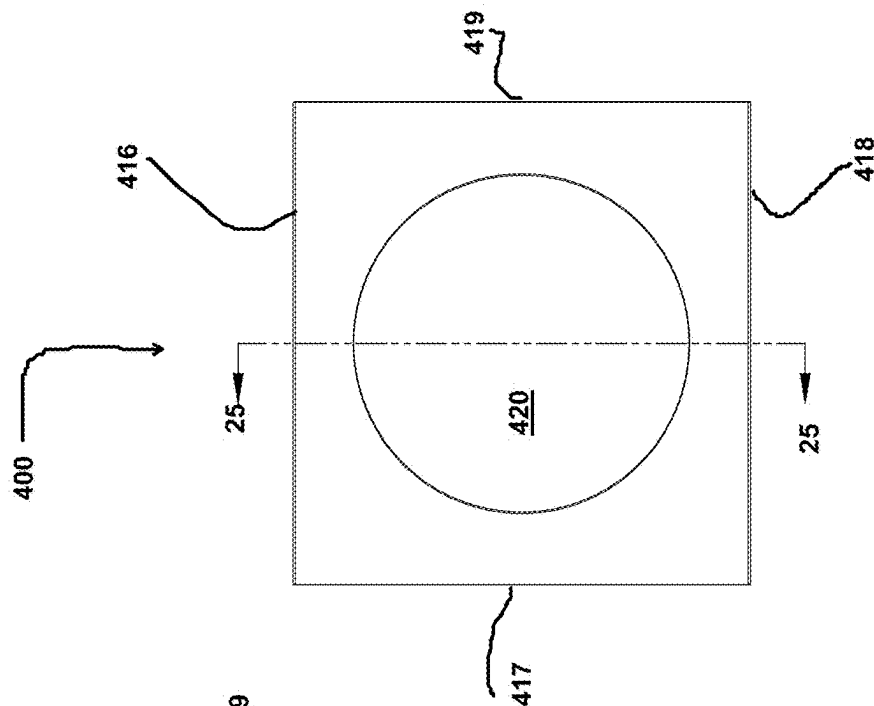


FIG. 24

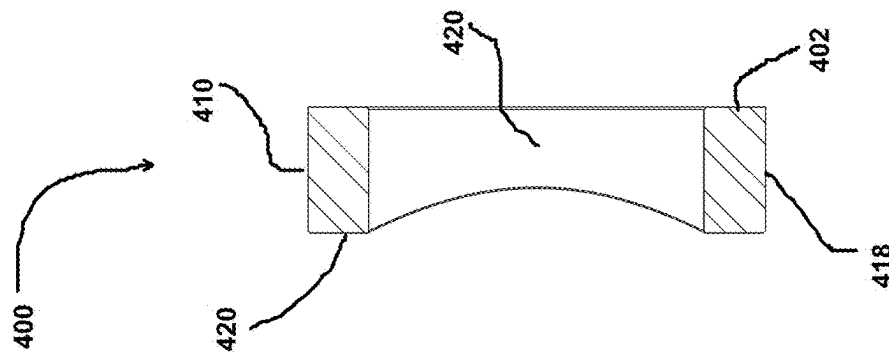


FIG. 25

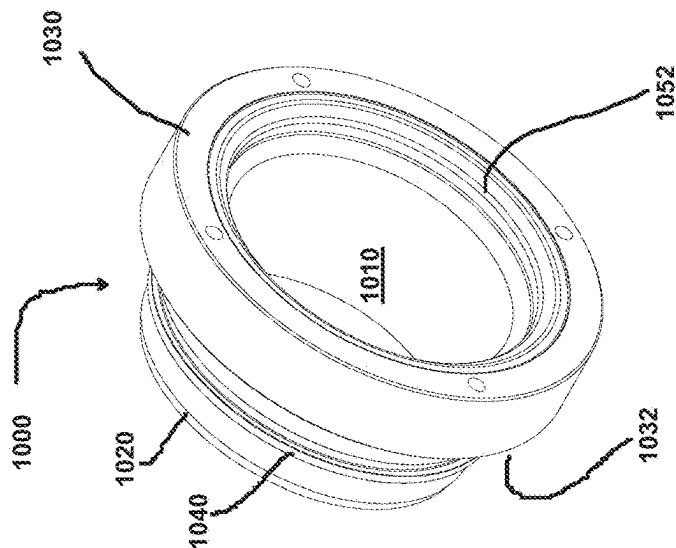


FIG. 26

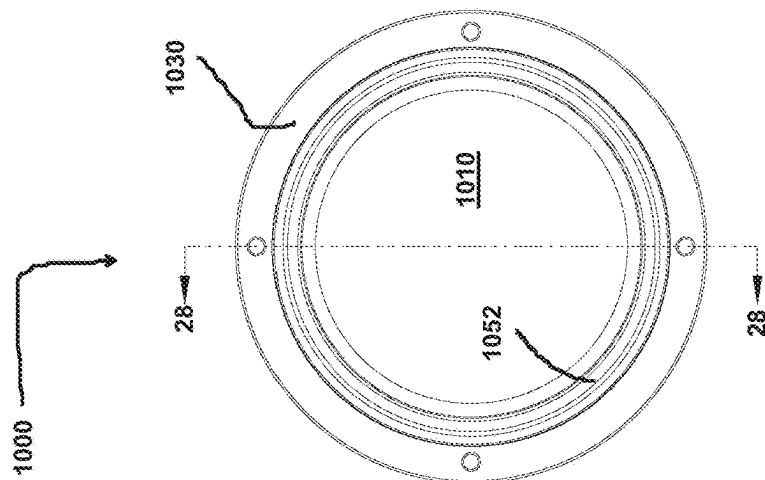


FIG. 27

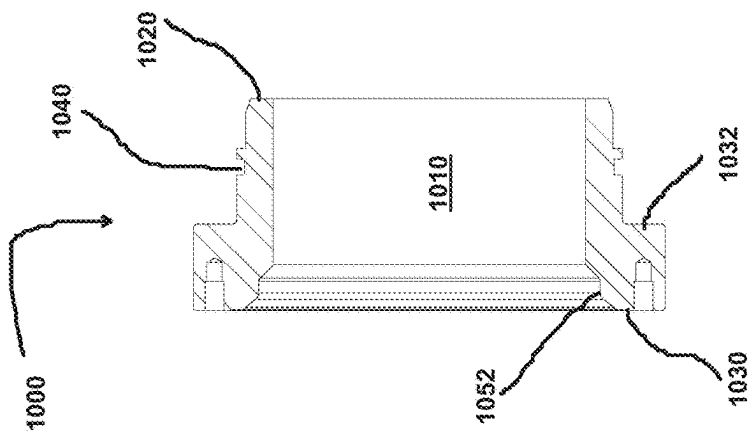
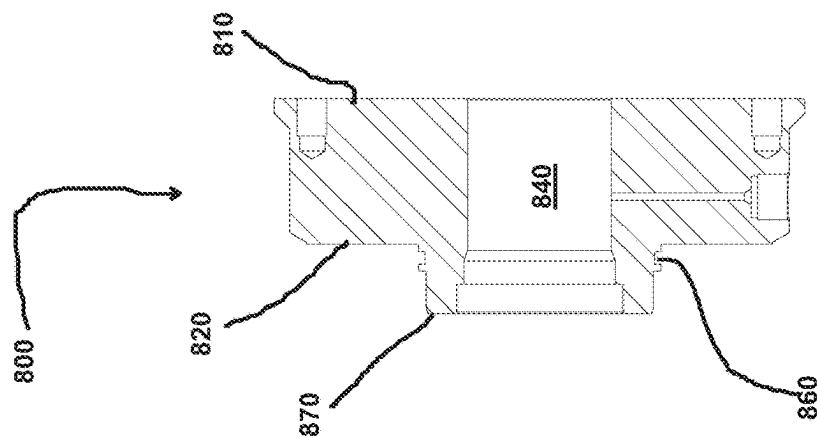
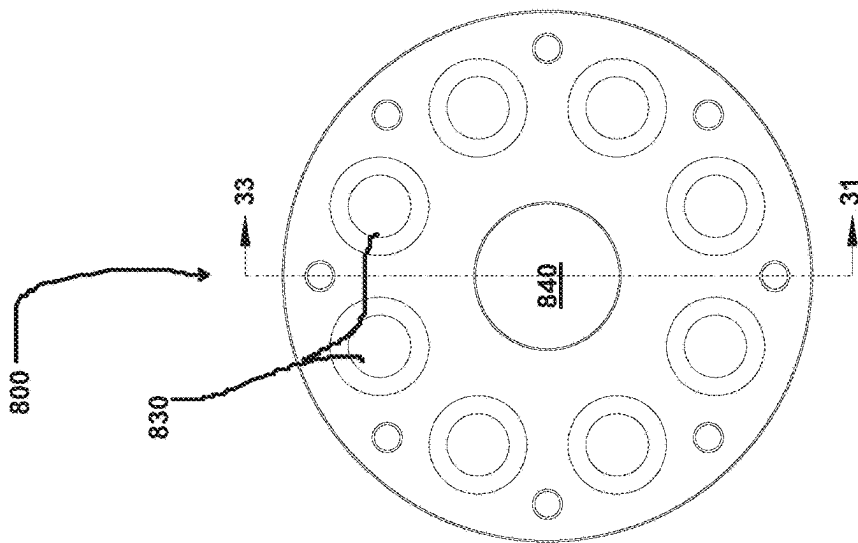
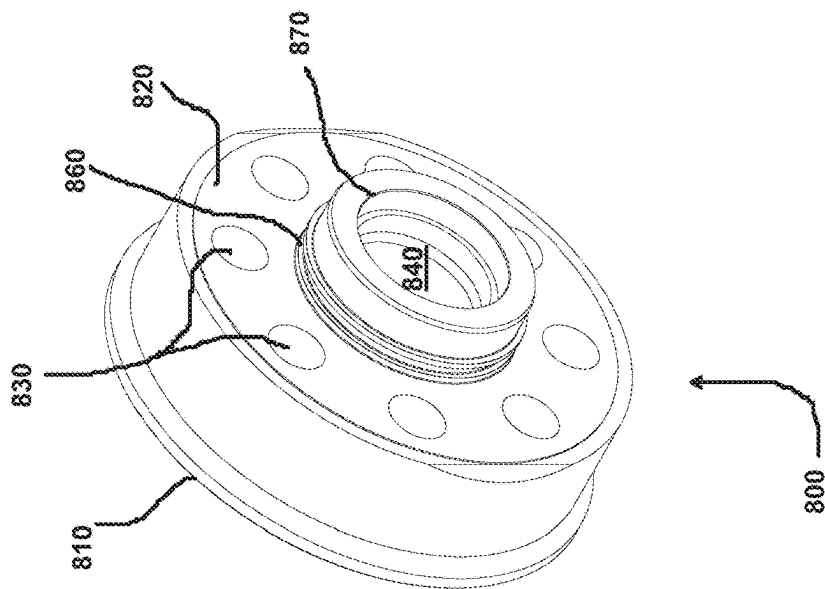
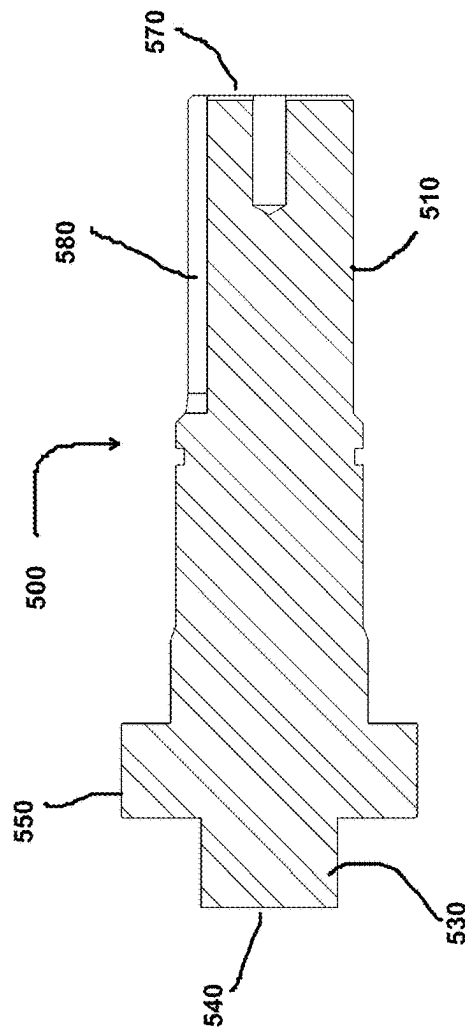
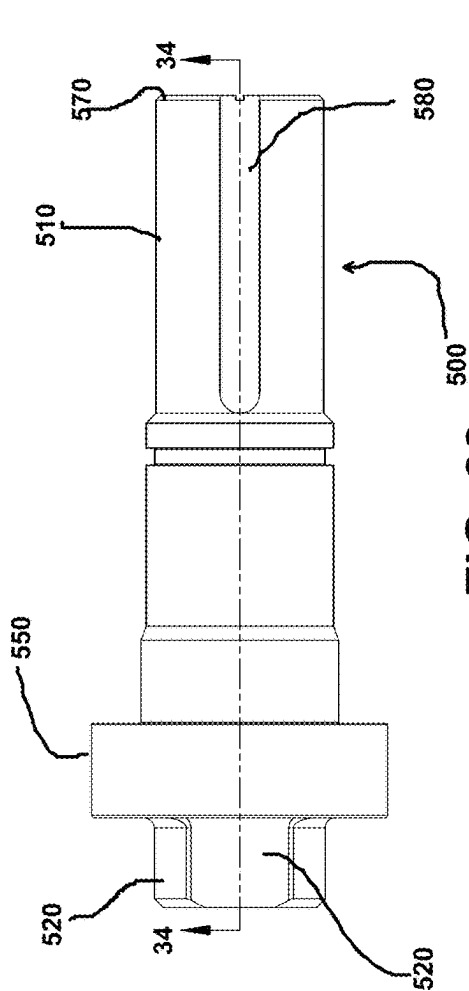
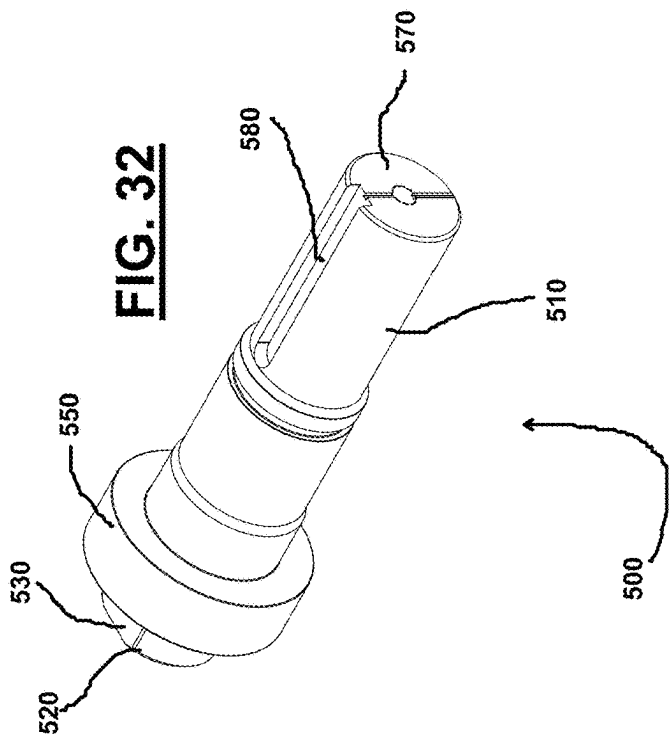


FIG. 28





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DOUBLE PIECE RETAINER FOR COMPACT VALVE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This is a continuation of U.S. patent application Ser. No. 17/857,231, filed Jul. 5, 2022 (issuing as U.S. Pat. No. 12,000,506 on Jun. 4, 2024), which is a continuation of U.S. patent application Ser. No. 16/931,957, filed Jul. 17, 2020 (issued as U.S. Pat. No. 11,378,190 on Jul. 5, 2022), which is a continuation of U.S. patent application Ser. No. 16/113,050, filed Aug. 27, 2018 (issued as U.S. Pat. No. 10,718,437 on Jul. 21, 2020), which is a continuation of U.S. patent application Ser. No. 14/631,970, filed Feb. 26, 2015 (issued as U.S. Pat. No. 10,060,538 on Aug. 28, 2018), which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/946,252, filed Feb. 28, 2014, all of which applications/patents are incorporated herein by reference and priority to/of such applications is hereby claimed.

BACKGROUND OF THE INVENTION

Conventionally available are side entry compact valves including a valve ball having integral opposed trunnions rotating in upper and lower trunnion support plates, which plates are supported by the valve body, with right and left valve seats and seals for sealing, and all of which components are inserted and removed from the valve body via one of the side openings of the valve, and which components are held in place during use with a threaded retainer.

The “compact” nature of compact valves generally results from a short or compact valve body having opposed “open” ends each defining a flange closure seat which is engaged by and establishes sealing with a flange closure which may be defined by a flange of a pipe section of the manifold or other piping assembly. Since no other body closure members are provided, the overall length of the valve body is short and compact as compared with the valve body structures of conventional ball valves. Accordingly, the compact manifold ball valves can be installed in piping systems such as flow control manifolds in offshore production platforms where minimal space is available.

One example of a type of side entry compact valve is disclosed in U.S. Pat. No. 6,669,171 for a “Compact Manifold Trunnion Ball Valve” which is incorporated herein by reference. The upper and lower trunnion support plates prevent the valve ball from transferring downstream loads to the seals and seats reducing frictional forces between the ball and seals.

However, in conventionally available side entry compact valves generally a single piece retainer is threadably connected to the valve body, wherein such single piece retainer must be rotated relative to both the valve body and the adjacent valve seat with such relative rotation causing wear to the seals between the valve seat and retainer along with the seals between the retainer and valve body.

It has been found that sealing elements in compact valves have a significant risk of becoming excessively worn and/or damaged during valve assembly and/or disassembly. It has been found that relative rotational movement between the valve components and the seals increase the risk of substantial wear and/or damage to the seals.

It would be advantageous to have a retainer both threadably connected to the valve body and concentrically posi-

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tioned in said body, wherein there is little no relative rotational movement between sealing elements for the valve seat and retainer.

It would be advantageous to have a retainer both threadably connected to the valve body and concentrically positioned in said body, wherein there is little no relative rotational movement between sealing elements for the valve body and retainer.

SUMMARY OF THE INVENTION

One embodiment generally relates to compact ball valves for use in conduit manifold systems. More particularly one embodiment includes a compact manifold ball valve having a valve body, valve ball, and valve stem; the valve body including a valve chamber having first and second ends, and a first flow passage intersecting the valve chamber and valve ball. The valve ball can be trunnion supported with at least one trunnion support element. A pair of movable seat assemblies can be used to seal the valve ball to the valve body. In one embodiment the above referenced components can be held in place during use with a threaded two piece retainer.

In one embodiment the two piece retainer longitudinally holds in place at least one trunnion support element restricting the amount of longitudinal movement of said trunnion support element. In one embodiment two trunnion support elements are longitudinally held in place with restricted to no longitudinal movement allowed.

In one embodiment is provided a retainer which is both threadably connected to the valve body and concentrically positioned in said body includes a sealing portion having little to no relative rotational movement between sealing elements for the valve seat and retainer section during valve assembly.

In one embodiment is provided a two piece retainer which is both threadably connected to the valve body and concentrically positioned in said body, wherein the two piece retainer includes first and second sections, wherein the first section having little to no relative rotational movement between sealing elements for the valve seat and retainer section during valve assembly.

In one embodiment is provided a two piece retainer which is both threadably connected to the valve body and concentrically positioned in said body, wherein the two piece retainer includes first and second sections, wherein the second section is threadably connected to the valve body such that rotation of the second section relative to the valve body causes either tightening or loosening of the first section relative to the valve body.

In one embodiment is provided a two piece retainer including first and second sections, wherein the second section is a ring, and rotational movement of the second section causes linear movement of the second section.

In one embodiment rotational movement of the first section relative to the valve body is constrained and/or prevented. In one embodiment at least one locking pin rotationally locks the first section relative to the valve body while allowing linear movement of the first section relative to the valve body.

In one embodiment rotational movement of the first section relative to the valve body is constrained and/or prevented while relative linear movement is allowed. In one embodiment the first section includes a detachable sealing element along with at least one locking pin that rotationally locks the first section relative to the valve body while

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simultaneously allowing linear movement of both the first section and sealing element relative to the valve body.

In one embodiment the first section includes a first detachable sealing element and the valve seat includes a second detachable sealing element. In this embodiment the first section also includes at least one locking pin that rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of the first section (and the first sealing element when attached to the first section) relative to the valve body, along with relative movement between the valve seat (and second sealing element when attached to the valve seat) with respect to the first section. In one embodiment the first sealing element forms a seal between the first section and valve body, while the second sealing element forms a seal between the first section and valve seat.

In one embodiment the first section includes first and second detachable and spaced apart sealing elements, along with at least one locking pin that rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of the first section (and the first and second sealing elements when attached to the first section) relative to the valve body. In one embodiment the first sealing element forms a seal between the first section and valve body, while the second sealing element forms a seal between the first section and valve seat.

In one embodiment the locking pins are have a relatively small shear force where a rotational force exceeding a predefined force will cause the shear pin(s) to shear and allow rotational movement between the first section and valve body.

In one embodiment two locking pins are used which are symmetrically spaced about the first section of the retainer.

In various embodiments at least 50 percent of the longitudinal length of the first retainer section enters the interior portion of the valve body. In various embodiments at least 55, 60, 65, 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, and 100 percent of the first retainer section enters the interior portion of the valve body during assembly. In various embodiments the amount of longitudinal length of the first retainer section entering the valve body during assembly is between about a range of any two of the above referenced percentages.

In various embodiments at least 50 percent of the longitudinal length of the second retainer section enters the interior portion of the valve body. In various embodiments at least 55, 60, 65, 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, and 100 percent of the second retainer section enters the interior portion of the valve body during assembly. In various embodiments the amount of longitudinal length of the second retainer section entering the valve body during assembly is between about a range of any two of the above referenced percentages.

In one embodiment the retainer pins are less than about 3 mm in diameter. In various embodiments the retainer pins are less than about 3, 2.5, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, and 0.1 mm. In various embodiments the diameters of the retainer pins are between about any two of the diameters specified in this paragraph.

In various embodiments the first retainer section includes a plurality of openings and the valve body includes a plurality of openings which match the plurality of openings for the first retainer section. In various embodiments a plurality of retainer pins can be placed in the plurality of openings to rotationally lock the first retainer section with respect to the valve body.

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In various embodiments the second retainer section locks in place the plurality of retainer pins in their respective plurality of openings.

While certain novel features of this invention shown and described below are pointed out in the annexed claims, the invention is not intended to be limited to the details specified, since a person of ordinary skill in the relevant art will understand that various omissions, modifications, substitutions and changes in the forms and details of the device illustrated and in its operation may be made without departing in any way from the spirit of the present invention. No feature of the invention is critical or essential unless it is expressly stated as being "critical" or "essential."

The drawings constitute a part of this specification and include exemplary embodiments to the invention, which may be embodied in various forms.

BRIEF DESCRIPTION OF DRAWINGS

For a further understanding of the nature, objects, and advantages of the present invention, reference should be had to the following detailed description, read in conjunction with the following drawings, wherein like reference numerals denote like elements and wherein:

FIG. 1 is a sectional side view of one embodiment the showing a compact valve body with a two piece retainer holding in position the valve ball and trunnion support elements along with the valve ball seats.

FIG. 2 is an end view of the compact valve shown in FIG. 1, but all valve components shown in an exploded condition from the second end of the body of the compact valve.

FIG. 3 is a sectional view of the compact valve shown in FIG. 4 taken along the lines 3-3.

FIG. 4 is a perspective view of the first retainer section shown in the valve of FIG. 1.

FIG. 5 is an end view of the first retainer section shown in FIG. 4.

FIG. 6 is a sectional view of the first retainer section shown in FIGS. 4 and 5, taken along the lines 6-6.

FIG. 7 is a perspective view of the second retainer section shown in the valve of FIG. 1.

FIG. 8 is an end view of the second retainer section shown in FIG. 7.

FIG. 9 is a sectional view of the second retainer section shown in FIGS. 7 and 8, taken along the lines 9-9.

FIG. 10 is a partially exploded perspective view of the valve of FIG. 1 with the two piece retainer of FIGS. 4 through 9 positioned to be installed in the valve body where the valve ball and seats are already installed in the valve body.

FIG. 11 is a sectional view of the partially exploded view of the embodiment shown in FIG. 10 taken along with lines 11-11.

FIGS. 12 through 15 schematically indicate the assembly steps of placement of the first and second retainer sections into the valve body.

FIG. 12 shows both the first and second retainer sections before placement in the valve body of the compact valve.

FIG. 13 shows the valve of FIG. 12 where the first retainer section has been partially inserted into the valve body, but the second retainer section has not yet been inserted into the valve body.

FIG. 14 shows the valve of FIG. 12 where both the first and second retainer sections have been partially inserted into the valve body.

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FIG. 15 shows the valve of FIG. 12 where both the first and second retainer sections have been completely inserted into the valve body.

FIG. 16 is a perspective view of the valve body taken from the first end of the valve body.

FIG. 17 is a perspective view of the valve body taken from the second end of the valve body.

FIG. 18 is a side view of the second end of the valve body.

FIG. 19 is a sectional view of the valve body taken along the lines 19-19.

FIG. 20 is a perspective view of the valve ball shown in the valve of FIG. 1.

FIG. 21 is an end view of the valve ball shown in FIG. 16.

FIG. 22 is a sectional view of the valve ball in FIGS. 20 and 21, taken along the lines 21-21.

FIG. 23 is a perspective view of the trunnion support element shown in the valve of FIG. 1;

FIG. 24 is an end view of the trunnion support element shown in FIG. 23.

FIG. 25 is a sectional view of the trunnion support element shown in FIGS. 23 and 24, taken along the lines 25-25.

FIG. 26 is a perspective view of the seat shown in the valve of FIG. 1.

FIG. 27 is an end view of the seat shown in FIG. 26.

FIG. 28 is a sectional view of the seat shown in FIGS. 26 and 27, taken along the lines 28-28.

FIG. 29 is a perspective view of the bonnet shown in the valve of FIG. 1.

FIG. 30 is an end view of the bonnet shown in FIG. 29.

FIG. 31 is a sectional view of the bonnet shown in FIGS. 29 and 30, taken along the lines 31-31.

FIG. 32 is a perspective view of the stem shown in the valve of FIG. 1.

FIG. 33 is a side view of the stem shown in FIG. 32.

FIG. 34 is a sectional view of the stem shown in FIGS. 32 and 33, taken along the lines 34-34.

DETAILED DESCRIPTION OF THE INVENTION

Detailed descriptions of one or more preferred embodiments are provided herein. It is to be understood, however, that the present invention may be embodied in various forms. Therefore, specific details disclosed herein are not to be interpreted as limiting, but rather as a basis for the claims and as a representative basis for teaching one skilled in the art to employ the present invention in any appropriate system, structure or manner.

Overview of Compact Valve With Two Piece Retainer

FIG. 1 is a sectional side view of one embodiment showing an assembled compact valve body 20 with a two piece retainer 1100 (first 1200 and second 1300 sections) installed in the valve body 20 retaining a valve ball 200 and seats 1000,1000' in the valve body 20. FIG. 2 is an end view of compact valve 10, showing certain valve components in an exploded condition—this end view being taken from the second end 40 of body 20. FIG. 3 is a sectional side view of the compact valve 10 taken along the lines 3-3 generally showing all valve components in an exploded condition.

FIG. 10 is a partially exploded perspective view of compact valve 10 showing valve body 20 with the two piece retainer 1100 (first 1200 and second 1300 sections) positioned to be installed in the valve body 20 where the valve ball 200 and seats 1000,1000' are already installed in the

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valve body 20. FIG. 11 is a sectional view through the partially exploded view of the compact valve 10 taken along with lines 11-11.

In one embodiment compact valve 10 can comprise body 20, valve ball 200, and two piece retainer 1100. Valve ball 200 can be rotatably and sealably connected to body 20. Valve body 20 can include flow passage 60. Valve ball 200 can include flow passage 270. To allow flow through flow passage 60 of valve body 20, flow passage 270 can be aligned, either partially or wholly, with flow passage 60. To restrict or prevent flow through flow passage 60, flow passage 270 can be non-aligned, either partially or wholly, with flow passage 60. Alignment of flow passage 270 can be made by rotating valve ball 200 relative to valve body 20.

Valve ball 200 can be rotatably supported in body 20 by trunnions 220 and 260. Trunnion 220 can be rotatably connected to trunnion support element 400. Trunnion 260 can be rotatably connected to trunnion support element 450. Valve ball 200 can be operatively connected to stem 500, such as through stem recess 220. Stem recess 230 can have a rectangular cross section, although other types of operable connections can be made such as square, ribbed, or any non-circular shaped recess; or any type of mechanical connection.

In one embodiment rotating stem 500, such as in the direction of arrow 1500, causes similar rotation of valve ball 200. Stem 500 itself can be rotatably connected to body 20. Trunnion support element 400 can be supported in body 20. Accordingly, although valve ball 200 can rotate relative to valve body, longitudinal movement (i.e., arrows 1510,1520) can be restricted and/or prevented depending on the relative amount of longitudinal movement allowable between: valve ball 200 and trunnion support elements 400,450; and trunnion support elements 400,450 and valve body 20. It is preferred that longitudinal movement between valve ball 200 and valve body 20 be minimized. It is also preferred that stem 500 be aligned with cylindrical opening 420 of trunnion support element 400 (and cylindrical opening 470 of second trunnion support element 450) as misalignment can cause difficulty in turning valve ball 200.

Valve ball 200 can be sealably connected to valve body 20 by means of opposed and biased seats 1000 and 1000'. Seats 1000 and 1000' can be supported in valve body 20 and biased toward valve ball 200, respectively in the directions of arrows 1510 and 1520. Accordingly, where flow passage 270 of valve ball 200 is rotated 90 degrees in relation to flow passage 60 of valve body 20 (e.g., in direction of arrow 1500), flow is cut off by the sealing effect of seats 1000 and 1000'. As the relative rotation between flow passage 270 of valve ball 200 and flow passage 60 of valve body 20 is reduced from 90 ninety degrees, the restriction of flow by valve ball 200 is reduced. Maximum flow through valve 10 is achieved when flow passage 270 of valve ball 200 is aligned with flow passage 60 of valve body 20 (i.e., when the relative rotation is zero degrees).

Generally two piece retainer 1100 can retain the valve components inside valve body 20. FIGS. 4 through 9 show various views of components of the two piece retainer 1100 which is generally comprised of rotationally static first section 1200 and rotating second section 1300.

FIG. 4 is a perspective view of first retainer section 1200. FIG. 5 is an end view of the first retainer section 1200. FIG. 6 is a sectional view of first retainer section 1200 taken along the lines 6-6. First section 1200 can include first and second shoulders 1250,1260 along with first and second circumferential areas 1252,1262 which accommodate seat 1000. Preferably, planar surface 1222 on second end limits the amount

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of longitudinal movement of first 400 and second 450 trunnion support elements which second retainer section 1300 is tightened in valve body 20. When retainer 1100 is installed in valve body 20, there exists a defined longitudinal length between planar surface 1222 of second retainer section 1200 and third shoulder 116 of valve body 20 so that trunnion support elements 400, 450 each have a limited defined space to sit in (i.e., the space between planar surface 1222 and third shoulder 116).

Preferably, a seal is maintained between first retainer section 1200 and valve body 20, which can be a lip seal. To facilitate this sealing, seal recess 1224 can be provided which can accommodate seals 1042, which seals can each include a lip seal and back-up ring (the back up ring increasing the sealing pressure rating and resisting extrusion of the lip seal). In one embodiment one or both of the back-up rings can be omitted.

Also preferably, a seal is maintained between first retainer section 1200 and valve seal 1000, which can be an o-ring type seal. To facilitate this sealing, seal recess 1040 can be provided which can accommodate seals 1042, which seals can each include an o-ring and back-up ring (the back up ring increasing the sealing pressure rating and resisting extrusion of the o-ring).

FIG. 7 is a perspective view the second retainer section 1300. FIG. 8 is an end view of second retainer section 1300. FIG. 9 is a sectional view of second retainer section 1300 taken along the lines 9-9. Second retainer section can be a threaded ring like section having first end 1310 and second end 1320 with a central opening 1305. A threaded area 1340 can be used to threadably connect second section 1300 to valve body 20. A plurality of spaced apart openings 1314 can be used to connect section retainer section 1300 to a tool for tightening or loosening of second retainer section relative to valve body 20.

Assembly and Disassembly of Valve With Two Piece Retainer

FIG. 1 is a sectional side view of the assembled compact valve 10 using two piece retainer 1100 to hold in place various internal valve components. FIG. 2 is an end view of the compact valve 10, but all valve components shown in an exploded condition from the second end 40 of valve body 20 of compact valve 10. FIG. 3 is a sectional view of exploded compact valve 10 taken along the lines 3-3.

FIGS. 10 through 15 schematically indicate the assembly steps of placement of the retainer 1100's first 1200 and second 1300 retainer sections into valve body 20.

FIG. 10 is a partially exploded perspective view of one embodiment of valve 10 showing compact valve body 20 with the two piece retainer 1100 positioned to be installed in valve body 20 where the valve ball 200, trunnion support elements 400, 450, and seats 1000,1000' are already installed in the valve body 20. FIG. 11 is a sectional view of the partially exploded view of valve 10 taken along with lines 11-11.

FIG. 12 shows both the first 1200 and second 1300 retainer sections before placement in valve body 20 of compact valve 10. Arrow 1400 schematically indicates the longitudinal movement which will occur in retain 1100 for pieces 1200,1300, although second retainer section 1300 which include rotational movement.

First section can include detachable seal 1226 which sits in recess 1224. Detachable seal 1226 can be a lip type seal. In one embodiment seal 1226 can include an extrusion ring to prevent extrusion of seal 1226. In one embodiment seal 1226 can be an o-ring type seal.

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Seat retainer 1000 section can include detachable seal 1042 which sits in recess 1040. Detachable seal 1042 can be various conventionally available seals. In one embodiment seal 1042 can include an extrusion ring to prevent extrusion of seal 1042. In one embodiment seal 1042 can be an o-ring type seal.

FIG. 13 shows valve 10 where first retainer section 1200 has been partially inserted into valve body 20, but second retainer section 1300 has not yet been inserted into valve body 20. At this point seal 1226 has not yet made contact with valve body 20. Seal 1224 is shown in the vicinity of threads 140. Also at this point seal 1042 of seat 1000 has not yet made contact with first retainer section 1200. Second retainer section 1300 is shown on the outside of valve body 20. Arrow 1410 schematically indicates that second retainer section 1300 will be rotated relative to valve body 20, however, during such rotation in the direction of arrow 1410 first retainer section will remain rotationally fixed relative to valve body 20 (and also relative to valve seat 1000). Arrow 1400 schematically indicates longitudinal movement of first retainer section 1200 relative to valve body 20 while simultaneously remaining rotationally static relative to valve body 20. In FIG. 13 is also shown retaining pins 1246,1246' in openings 1247 of first retainer 1200. Retainer pin 1246 will seat in opening 180 for restricting rotational movement of first retainer section 1200 relative to valve body 20. Retaining pin 1246' will seat in opening 182 for restricting rotational movement of first retainer section 1200 relative to valve body 20 (rotational movement schematically indicated by arrow 1410). However, both retaining pins 1246,1246' allow linear movement of first retainer section 1200 relative to valve body 20 (schematically indicated by arrow 1400).

FIG. 14 shows valve 10 where both first 1200 and second 1300 retainer sections have been partially inserted into valve body 20. Threads 1340 of second retainer section 1300 have now engaged threads 140 of valve body. Seal 1226 has engaged surface 132 of valve body 20. Seal 1042 has engaged surface 1262 of first retainer section 1200. Both retaining pins 1246,1246' allow linear movement of first retainer section 1200 relative to valve body 20 (schematically indicated by arrow 1400). Rotational movement (schematically indicated by arrow 1410) of second retainer section 1300, with its threads 1340 engaging threads 140 of valve body 20, will cause second retainer section 1300 to move in the direction of arrow 1400 and push first retainer section 1200 in the direction of arrow 1400.

FIG. 15 shows valve 10 where both first 1200 and second 1300 retainer sections have been completely inserted into valve body 20. Continued rotational movement of second retainer section 1300 compared to FIG. 14 (schematically indicated by arrow 1410), with second retainer 1300 threads 1340 engaging threads 140 of valve body 20, will cause second retainer section 1300 to continue to also move linearly in the direction of arrow 1400 and continue to push first retainer section 1200 linearly in the direction of arrow 1400. During the linear pushing by second retainer section 1300, first retainer section 1200 moves only linearly (schematically indicated by arrow 1400) and is rotationally constrained by pins 1246 and 1246'. In this manner the seals 1226 and 1042 only see relative linear movement and no relative rotational movement. Eventually, second retainer section 1300 will push first retainer section 1200 enough in the direction of arrow 1400 that second end 1220 will contact both first 400 and second 450 trunnion supports locking linearly in place these trunnion support elements with shoulder 116 of valve body 20.

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Disassembly of valve **20** can be performed by rotating second retainer section **1300** in a direction opposite of the rotational direction of arrow **1410**. During disassembly, second retainer section **1200** also is constrained from rotational movement by pins **1246** and **1246'**.

Valve Components

Below various individual components of valve **10** will be reviewed.

FIG. **16** is a perspective view of valve body **20** taken from the first end **30**. FIG. **17** is a perspective view of valve body **20** taken from second end **40**. FIG. **18** is a side view of valve body **20** taken from the second end **40**. FIG. **19** is a sectional view of valve body **20** taken along the lines **19-19**. Valve body can include first end **30** and second end **40**. Flow passage **60** can proceed from first end **30** through second end **40**. Stem **500** can be rotatably connected to valve body **20**. Valve body **20** can include internal chamber **50** where valve ball **200** remains during use. From first end **30** to second end **40**, valve body **20** can include threaded area **140**, second cylindrical area **135**, second shoulder **134**, first cylindrical area **132**, first shoulder **130**, internal bore **50**, shoulder **116**, cylindrical area **112**, shoulder **110**, cylindrical area **102**, cylindrical bore **102**. First and second shoulders **100,110** are found in seat recess **120**. Bonnet recess **80** can be provided for allowing attachment of bonnet **800** to valve body **20**. One or more threaded bores **88** can be provided for fasteners **832**. Openings **834,834'** can be used for inserting a stop pin **836**. Lubrication port **160** and vent/bleeding port **162** can be respectively provided for lubrication fitting **161** and vent/bleeding fitting **163**.

FIGS. **29** through **34** show the components of the valve bonnet **800** and stem **700** assembly.

FIG. **32** is a perspective view of stem **500**. FIG. **33** is a side view of stem **500**. FIG. **34** is a sectional view of stem **500** taken along the lines **34-34**. Stem **500** can comprise shaft **510**, and ball drive element **520**. Ball drive element **520** can include substantially planar drive surface **530**. Circumferential bearing surface **550** can be included for rotatably connecting stem **500** with valve body **20** through stem receptacle **70**. Shaft **510** can include recess **580** for a key **582** which key can be used to operatively connect stem **500** to an actuator.

FIG. **29** is a perspective view of bonnet **800**. FIG. **30** is an end view of bonnet **800**. FIG. **31** is a sectional view of bonnet **800** taken along the lines **31-31**. Bonnet **800** can be used to prevent stem **500** from blowing out of valve body **20**. Bonnet **800** can be connected to valve body **20** through one or more fasteners **832**. Bonnet **800** can include tip **870**. Preferably, a seal is maintained between bonnet **800** and valve body **20**. To facilitate this seal, recess **860** can be provided which can include seal **862** such as an o-ring and back-up ring (the back up ring increasing the sealing pressure rating and resisting extrusion of the o-ring).

FIG. **20** is a perspective view of valve ball **200**. FIG. **21** is an end view of valve ball **200**. FIG. **22** is a sectional view of valve ball **200** taken along the lines **21-21**. Valve ball **200** can be spherically shaped and include top **210**, bottom **250**, and flow passage **270**. Upper spherical surface segment can be located by top **210** and lower spherical surface segment **310** can be located by bottom **250**. Stem recess **230** can be included.

Trunnion **220** can be located on top **210** and can include cylindrical area **222**. Cylindrical area **222** can be rotatably or pivotally connected to trunnion support element **400**. Bearing surface **224** can be substantially planar and slidingly contact first surface **402** of trunnion support element **400**. Preferably, a trunnion shim **404** is provided to act as a

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bearing surface between bearing area **224** of valve ball **200** and trunnion support element **400**. Also preferably, trunnion bearing **425** is provided between trunnion **220** and trunnion support element **400**. Also preferably, trunnion **220** is prevented from contacting valve body **20** to prevent wear.

Trunnion **260** can be located on bottom **250** and can include cylindrical area **262**. Cylindrical area **262** can be rotatably or pivotally connected to trunnion support element **450**. Bearing surface **264** can be substantially planar and slidingly contact first surface **452** of trunnion support element **450**. Preferably, a trunnion shim **454** is provided to act as a bearing surface between bearing area **264** of valve ball **200** and trunnion support element **450**. Also preferably, trunnion bearing **475** is provided between trunnion **260** and trunnion support element **450**. Also preferably, trunnion **260** is prevented from contacting valve body **20** to prevent wear.

Upper and lower spherical segments **300,310** can each be sealably connected to both seats **1000,1000'**.

FIG. **23** is a perspective view of trunnion support element **400** shown in the valve of FIG. **1**. FIG. **24** is an end view of trunnion support element **400**. FIG. **25** is a sectional view of trunnion support element **400**, taken along the lines **25-25**. Trunnion support element **450** can be substantially similar to trunnion support element **400** and will not be described separately. Cylindrical opening **420** can be provided to slidably connect upper trunnion **220** of valve ball **200**. First and third sides **416,418** of trunnion support element **400** can be held in place between third shoulder **116** of valve body **20** and second end **1220** of first retainer section **1200** (limiting movement of trunnion support element **400** in the direction of arrows **1510,1520**).

FIG. **26** is a perspective view of **1000**. FIG. **27** is an end view of seat **1000**. FIG. **28** is a sectional view of seat **1000** taken along the lines **28-28**. Seat **1000'** can be constructed substantially similar to seat **1000** (and seat **1000'** will not be individually described). Seat **1000** can include internal passage **1010**, first side **1020**, second side **1030**, along with first and second circumferential areas **1070,1080**. First and second circumferential areas **1070, 1080** of seat **1000** sit respectively in first and second circumferential areas **1252, 1262** of retainer **1200**. (For seat **1000'**, first and second circumferential areas **1070,1080** sit respectively in cylindrical areas **112,102** of valve body **20**).

Preferably, a seal is maintained between seat **1000** and valve body **20** (or second retainer section **1200**). To facilitate this seal, recess **1040** can be provided which can include seal **1042**, which seal can each include an o-ring and back-up ring (the back up ring increasing the sealing pressure rating and resisting extrusion of the o-ring). Preferably, a seal is maintained between seat **1000** and valve ball **200**. To facilitate this seal, recess **1050** can be provided which can include seal **1052**, which seal **1052** seals with upper and lower spherical surfaces **300,310** of valve ball **200** (seat **1000'** is substantially the same). Seal **1052** can be installed in recess **1050** using any conventional means, such as crimping, adhesive, friction fit, or other means.

To facilitate sealing at lower line pressures between seats **1000,1000'** and valve ball **200**, biasing members **1090,1090'** can be provided which respectively push seat **1000** in the direction of arrow **1520** and seat **1000'** in the direction of arrow **1510** (both seats toward valve ball **200**). At higher line pressures seats **1000,1000'** will tend to be pushed toward valve ball **200** because the larger exposed surface area on first side **1020** as opposed to second side **1030**. This is because line fluid will not be able to pass seal **1052** thereby limiting the amount of surface area to which the line fluid can assert pressure. In an alternative embodiment a second

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set of biasing members can be provided for second circumferential area **1080** (or area **1080'** for seat **1000'**). In one embodiment, peripheral groove **1082** can be provided to reduce the amount of surface area of seat **1000** which contacts valve body **20** (or retainer **1200** for seat **1000'**) thereby reducing frictional forces between these items.

In any of the embodiments a lip seal can be used as the sealing element. In any of the embodiments an extrusion ring can be used in combination with the lip seal.

The following is a list of reference numerals:

LIST FOR REFERENCE NUMERALS	
(Reference No.)	(Description)
10	compact manifold ball valve
20	body
30	first end
38	bores for fasteners
40	second end
42	raised area
44	peripheral groove for seal
46	seal
48	bores for fasteners
50	internal chamber
60	flow passage
70	stem receptacle
72	bore
80	bonnet recess
88	bores for fasteners
90	seat recess
100	first shoulder
102	cylindrical area
110	second shoulder
112	cylindrical area
116	third shoulder
118	cylindrical area
120	seat recess
130	first shoulder
132	cylindrical area
134	second shoulder
135	cylindrical area
138	third shoulder
139	cylindrical area
140	threaded area
150	internal bore
152	cylindrical area
160	lubrication port
161	lubrication fitting
162	vent/bleeding port
163	vent/bleeding fitting
180	opening
182	opening
200	valve ball
210	top
220	trunnion
222	cylindrical area
224	bearing area
230	recess for stem
250	bottom
260	trunnion
262	cylindrical area
264	bearing area
270	flow passage
280	first end
290	second end
300	spherical surface segment
310	spherical surface segment
400	first trunnion support element
402	first surface
404	trunnion bearing
405	trunnion shim/bearing
410	second surface
416	first side
417	second side
418	third side
419	fourth side
420	cylindrical opening

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-continued

LIST FOR REFERENCE NUMERALS	
(Reference No.)	(Description)
425	trunnion bearing
430	recessed area
450	second trunnion support element
452	first surface
454	trunnion bearing
455	trunnion shim/bearing
460	second surface
466	first side
467	second side
468	third side
469	fourth side
470	cylindrical opening
475	trunnion bearing
480	recessed area
500	stem
510	shaft
520	ball drive element
530	substantially planar drive surface
540	tip
550	bearing surface
552	bearing
570	top
580	recess for key
582	key
800	bonnet
810	first end
820	second end
830	plurality of recessed bores
832	plurality of fasteners
833	plurality of caps
834	opening for stop pin
836	stop pin
840	opening
850	recess
852	seal
860	recess
862	seal
870	tip
880	flat area
890	bore for stop pin
892	stop pin
894	grease bore
896	lubrication fitting
1000	seat
1010	internal passage
1020	first side
1022	planar surface
1030	second side
1032	shoulder
1040	recess
1042	seal and extrusion ring
1044	seal and extrusion ring
1052	seal or face seal
1090	circular wave spring
1100	two piece retainer
1200	first piece of retainer
1210	first end
1212	planar surface
1214	first recessed area
1220	second end
1222	planar surface
1224	seal recess
1226	seal
1230	internal passage
1242	first end
1244	second end
1246	plurality of shearable retainers to restrict rotation
1247	plurality of openings
1248	peripheral edge
1250	first shoulder
1252	first circumferential area
1260	second shoulder
1262	second circumferential area
1270	third shoulder
1272	third circumferential area

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-continued

LIST FOR REFERENCE NUMERALS	
(Reference No.)	(Description)
1300	second piece of retainer
1305	opening
1310	first end
1314	plurality of openings
1320	second end
1330	internal passage
1340	threaded area
1400	arrow
1410	arrow
1500	arrow
1510	arrow
1520	arrow

All measurements disclosed herein are at standard temperature and pressure, at sea level on Earth, unless indicated otherwise. All materials used or intended to be used in a human being are biocompatible, unless indicated otherwise.

It will be understood that each of the elements described above, or two or more together may also find a useful application in other types of methods differing from the type described above. Without further analysis, the foregoing will so fully reveal the gist of the present invention that others can, by applying current knowledge, readily adapt it for various applications without omitting features that, from the standpoint of prior art, fairly constitute essential characteristics of the generic or specific aspects of this invention set forth in the appended claims. The foregoing embodiments are presented by way of example only; the scope of the present invention is to be limited only by the following claims.

We claim:

1. A compact manifold ball valve, comprising:

- (a) a valve body, the valve body having first and second ends, a valve passage having a longitudinal axis;
- (b) a valve stem, the valve stem being rotatably operatively to the valve body;
- (c) a valve ball, the valve ball being located in the valve passage, the valve ball having an upper end, lower end, and flow passage, and a pair of trunnions;
- (d) a pair of trunnion support plates, the pair of trunnion support plates being located in the valve chamber and rotatably connected to the pair of trunnions;
- (e) a two piece retainer, the two piece retainer being detachably connected to the valve body and limiting an amount of longitudinal movement of at least one of the pair of trunnion support plates;
- (f) the two piece retainer having first and second sections, the first section being sized and shaped to be installed in the valve passage with a generally longitudinal movement, the first section having a first annular face that engages both of said pair of trunnion support plates and a first annular surface spaced from said first annular face, wherein the first section is connected to the valve body such that the first section is prevented from rotating relative to the valve body;
- (g) the second section being a ring member being detachably connected into the valve passage of the valve body forming a detachable connection;
- (h) wherein the second section is positioned to push on the first section responsive to a formation of said detachable connection of the ring member to the valve body;

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- (i) wherein rotation of the second section relative to the valve body enables selective tightening or loosening of the first section relative to the valve body;
- (j) wherein the second section is a ring, and rotational movement of the second section causes linear movement of the second section relative to the valve body;
- (k) wherein rotational movement of the first section of the two piece retainer relative to the valve body is constrained; and
- (l) wherein at least one locking pin between the first section and the valve body rotationally locks the first section relative to the valve body while allowing linear movement of the first section relative to the valve body, and the second section is in direct contact with the at least one locking pin.

2. The compact manifold ball valve of claim 1, wherein rotational movement of the first section relative to the valve body is prevented by a plurality of locking pins between the first section and the valve body, while relative linear movement is allowed between the first section and the valve body.

3. The compact manifold ball valve of claim 1, wherein the first section includes a detachable sealing element along with the at least one locking pin between the first section and the valve body rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of both the first section and the detachable sealing element relative to the valve body.

4. The compact manifold ball valve of claim 1, wherein the first section includes a first detachable sealing element which fluidly seals the first section to the valve body.

5. The compact manifold ball valve of claim 4, wherein the first and second detachable sealing elements are spaced apart and attached to the first section, and the at least one locking pin rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of the first section and the attached first and second detachable sealing elements relative to the valve body.

6. The compact manifold ball valve of claim 4, wherein the at least one locking pin rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of the first section and the first detachable sealing element relative to the valve body, along with relative movement between the first section and the second detachable sealing element attached to the first valve seat with respect to the first section.

7. The compact manifold ball valve of claim 6, wherein the at least one locking pin between the first section and the valve body has a predefined shearing force where a rotational force exceeding the predefined shearing force will cause the at least one locking pin to shear and allow relative rotational movement between the first section and valve body.

8. The compact manifold ball valve of claim 1, further including first and second valve seats, wherein the first valve seat is operatively connected to a first biasing member and the second valve seat is operatively connected to a second biasing member, wherein the first and second biasing members respectively bias their respective valve seat to maintain contact with the valve ball.

9. The compact manifold ball valve of claim 8, wherein the two piece retainer longitudinally holds in place at least one of the pair of trunnion support plates restricting the amount of longitudinal movement of said at least one of said pair of trunnion support plates.

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10. The compact manifold ball valve of claim 9, wherein the two piece retainer longitudinally holds in place both of said pair of trunnion support plates such that no longitudinal movement is allowed.

11. The compact manifold ball valve of claim 9, wherein the first section includes a sealing portion having no relative rotational movement between sealing elements for the first valve seat during relative movement between the first section and first valve seat during valve assembly, and little to no relative rotational movement between sealing elements for the valve body and two piece retainer during relative movement between the first section and valve body during valve assembly.

12. The compact manifold ball valve of claim 9, wherein the first section includes a sealing portion having no relative rotational movement between sealing elements for the first valve seat during relative movement between the first section and first valve seat during valve assembly.

13. The compact manifold ball valve of claim 12, wherein the second section is both detachably connected to the valve body and concentrically positioned in said valve body relative to said longitudinal axis.

14. A compact manifold ball valve, comprising:

- (a) a valve body, the valve body having first and second ends, a valve passage having a longitudinal axis;
- (b) a valve stem, the valve stem being rotatably connected to the valve body;
- (c) a valve ball, the valve ball being located in the valve passage, the valve ball having an upper end, lower end, and flow passage, and a pair of trunnions, wherein the valve ball is connected to and rotates with the stem;
- (d) a pair of spaced apart trunnion support plates being located in the valve chamber and rotatably connected to the pair of trunnions;
- (e) a two piece retainer being detachably connected to the valve body and limiting the amount of longitudinal movement of the pair of spaced apart trunnion support plates;
- (f) the two piece retainer having first and second sections, the first section being sized and shaped to be installed in the valve passage, the first section having a first annular face that engages both of said pair of spaced apart trunnion support plates and an annular surface spaced from said first annular face, wherein the first section is connected to the valve body such that the first section is prevented from rotating relative to the valve body;
- (g) the second section being a ring member detachably connectable into the valve passage of the valve body and forming a detachable connection with the valve body;

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(h) wherein the second section is positioned to push on the first section responsive to a formation of said detachable connection; and

(i) wherein rotation of the second section relative to the said valve body enables selective tightening or loosening of the first section relative to the said valve body, and the first section includes at least one locking pin spanning between the first section and the valve body, and the second section is in contact with the at least one locking pin.

15. The compact manifold ball valve of claim 14, wherein the first section includes a sealing portion having no relative rotational movement between sealing elements for a first valve seat during relative movement between the first section and first valve seat during valve assembly, and little to no relative rotational movement between sealing elements for the valve body and two piece retainer during relative movement between the first section and valve body during valve assembly.

16. The compact manifold ball valve of claim 14, wherein the second section is a ring, and rotational movement of the second section causes linear movement of the second section relative to the valve body.

17. The compact manifold ball valve of claim 14, wherein the at least one locking pin rotationally locks the first section relative to the valve body by a plurality of locking pins between the first section and the valve body, while allowing linear movement of the first section relative to the valve body.

18. The compact manifold ball valve of claim 14, wherein the at least one locking pin between the first section and the valve body has a predefined shearing force where a rotational force exceeding the predefined shearing force will cause the said at least one locking pin to shear and allow relative rotational movement between the first section and valve body.

19. The compact manifold ball valve of claim 14, wherein the first section includes a detachable sealing element along with the at least one locking pin between the first section and the valve body that rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of both the first section and the detachable sealing element relative to the valve body.

20. The compact manifold ball valve of claim 14, wherein the at least one locking pin rotationally locks the first section relative to the valve body while simultaneously allowing linear movement of the first section and a first sealing element relative to the valve body, along with relative movement between the first section and a second sealing element attached to the first valve seat with respect to the first section.

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